

Ans. to Q.No. 1(a)

$$G(s) = \frac{30}{s^5 + 8s^4 + 9s^3 + 6s^2 + s + 30}$$

For a transfer function of the form:
$$G(s) = \frac{b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

We can use canonical form.
since $n=5, a_4=8, a_3=9, a_2=6, a_1=1, a_0=30$ and $b_0=30$

state variables:
$$x_1 = y, x_2 = \dot{y}, x_3 = \ddot{y}, x_4 = y^{(3)}, x_5 = y^{(4)}$$

A, B, C, D Matrices (canonical form)
$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -30 & -1 & -6 & -9 & -8 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C = [30 \quad 0 \quad 0 \quad 0 \quad 0]$$

$$D = [0]$$

state equations:
$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

we can define
$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= x_5 \\ \dot{x}_5 &= -30x_1 - x_2 - 6x_3 - 9x_4 - 8x_5 + u \\ y &= 30x_1 \end{aligned}$$

Ans. to the Q. No. 1(b)

Given a state space system (A, B, C, D) , the transfer function is: $G(s) = C(sI - A)^{-1}B + D$

The state space equation in the Laplace domain:
$$sX(s) = AX(s) + BU(s) \dots (i)$$

$$Y(s) = CX(s) + DU(s) \dots (ii)$$

Solving (i) $\Rightarrow (sI - A)X(s) = BU(s)$

$$X(s) = (sI - A)^{-1}BU(s)$$

Using (i) into eqn (ii) $\Rightarrow$

$$Y(s) = C(sI - A)^{-1}BU(s) + DU(s)$$

$$\therefore G(s) = \frac{Y(s)}{U(s)} = \frac{C(sI - A)^{-1}BU(s) + DU(s)}{U(s)}$$

$$\therefore G(s) = C(sI - A)^{-1}B + D$$

Verifying Q1(a) using $G(s) = C(sI - A)^{-1}B + D$

From 1(a) $C = [30 \quad 0 \quad 0 \quad 0 \quad 0]$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -30 & -1 & -6 & -9 & -8 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$D = [0]$$

$$\therefore G(s) = [30 \quad 0 \quad 0 \quad 0 \quad 0] \left(\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -30 & -1 & -6 & -9 & -8 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + [0]$$

Using $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$ and computing matrix multiplication and verifying

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With MATLAB code, $G(s) = \frac{30}{s^5 + 8s^4 + 9s^3 + 6s^2 + s + 30}$

The formula works.